

Euclid's Elements — Book V

Proposition 16

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If four magnitudes are proportional, then they are also proportional alternately.

1 Introduction

Proposition V.16 is the alternando theorem.

From

$$a : b = c : d,$$

it derives

$$a : c = b : d.$$

This is the proposition that originally motivated the present Book V reconstruction. In the classical proof of Euclid XI.23, alternando is used to transform a proportion obtained from similar triangles into a form from which an inequality of corresponding magnitudes can be read.

The present formal proof preserves the Euclidean proposition-level dependency graph. It uses V.15 to pass to arbitrary positive multiples, V.11 to compose equal ratios, and V.14 to transfer comparison across those scaled equal ratios.

The production file is

```
CGJteamLab/Proposition5_16.lean
```

and it imports

```
CGJteamLab.Proposition5_11  
CGJteamLab.Proposition5_14  
CGJteamLab.Proposition5_15
```

so that the mathematical dependency graph is explicit in the module structure.

2 Euclid's Alternando Statement

The starting hypothesis is

$$a : b = c : d.$$

The desired conclusion is

$$a : c = b : d.$$

To verify the conclusion by Definition 5, we must fix arbitrary positive multiple indices m, n and prove that the three comparisons between

$$M_m(a) \quad \text{and} \quad M_n(c)$$

agree with the corresponding comparisons between

$$M_m(b) \quad \text{and} \quad M_n(d).$$

This is exactly the form supplied by V.14 once an appropriate scaled proportion has been constructed.

3 Step 1: Scale the First Ratio by V.15

For arbitrary m, n , V.15 gives

$$a : b = M_m(a) : M_m(b).$$

In Lean:

```
have hScaleAB :
  HilbertEudoxusProportion
    Geo
    a b
    (hilbertPositiveSegmentMultiple Geo m a)
    (hilbertPositiveSegmentMultiple Geo m b) :=
  euclid_proposition_5_15
  Geo a b m
```

This proportion is then reversed so that the scaled ratio appears first.

4 Step 2: Scale the Second Ratio by V.15

Similarly,

$$c : d = M_n(c) : M_n(d).$$

In Lean:

```
have hScaleCD :  
  HilbertEudoxusProportion  
    Geo  
    c d  
    (hilbertPositiveSegmentMultiple Geo n c)  
    (hilbertPositiveSegmentMultiple Geo n d) :=  
  euclid_proposition_5_15  
    Geo c d n
```

The two arbitrary indices m, n are independent, exactly as required by Definition 5 for the alternated proportion.

5 Step 3: Compose Ratios by V.11

The original proportion is

$$a : b = c : d.$$

After reversing the first V.15 equality we have

$$M_m(a) : M_m(b) = a : b.$$

V.11 composes this with the original proportion to obtain

$$M_m(a) : M_m(b) = c : d.$$

A second use of V.11 with

$$c : d = M_n(c) : M_n(d)$$

gives the fully scaled equality

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

This is the central bridge of the proof.

6 Step 4: Apply V.14

Now V.14 applies to the scaled equality

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

It yields the three comparison equivalences

$$M_m(a) < M_n(c) \iff M_m(b) < M_n(d),$$

$$M_m(a) = M_n(c) \iff M_m(b) = M_n(d),$$

and

$$M_n(c) < M_m(a) \iff M_n(d) < M_m(b).$$

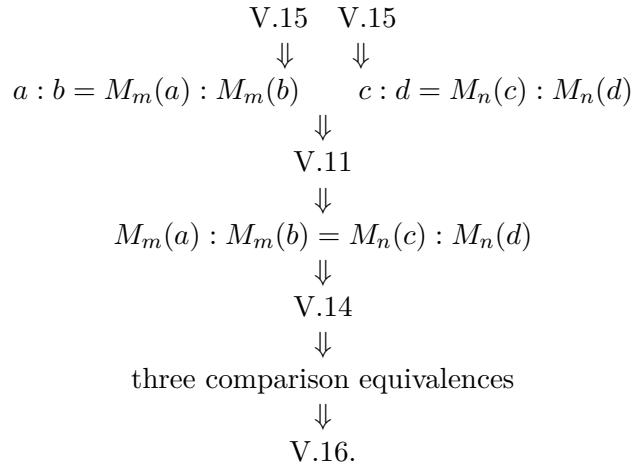
But these are exactly the three clauses of Definition 5 for

$$a : c = b : d.$$

Since m, n were arbitrary, alternando follows.

7 Proposition-Level Dependency DAG

The complete Euclid-numbered graph is



The original hypothesis

$$a : b = c : d$$

enters at the V.11 composition stage.

8 Formal Statement

The production theorem is:

```
theorem euclid_proposition_5_16
  [HilbertIncidence Geo]
  [HilbertArchimedeanPlane Geo]
  (a b c d : HilbertPositiveSegmentClass Geo)
  (hProp :
    HilbertEudoxusProportion Geo a b c d) :
  HilbertEudoxusProportion Geo a c b d := by

intro m n

have hScaleAB :=
  euclid_proposition_5_15
  Geo a b m

have hScaleCD :=
  euclid_proposition_5_15
  Geo c d n

have hScaledAB_to_CD :=
  euclid_proposition_5_11
  Geo
  (hilbertPositiveSegmentMultiple Geo m a)
  (hilbertPositiveSegmentMultiple Geo m b)
  a b
  c d
  (hilbertEudoxusProportion_symm
    Geo
    a b
    (hilbertPositiveSegmentMultiple Geo m a)
    (hilbertPositiveSegmentMultiple Geo m b)
    hScaleAB)
  hProp

have hScaled :=
  euclid_proposition_5_11
  Geo
  (hilbertPositiveSegmentMultiple Geo m a)
  (hilbertPositiveSegmentMultiple Geo m b)
  c d
  (hilbertPositiveSegmentMultiple Geo n c)
  (hilbertPositiveSegmentMultiple Geo n d)
  hScaledAB_to_CD
  hScaleCD

constructor
case left =>
  exact
    hilbertEudoxus_v14_lt_iff
    Geo
    (hilbertPositiveSegmentMultiple Geo m a)
    (hilbertPositiveSegmentMultiple Geo m b)
    (hilbertPositiveSegmentMultiple Geo n c)
    (hilbertPositiveSegmentMultiple Geo n d)
```

```

      hScaled

case right =>
  constructor
case left =>
  exact
    hilbertEudoxus_v14_eq_iff
      Geo
      (hilbertPositiveSegmentMultiple Geo m a)
      (hilbertPositiveSegmentMultiple Geo m b)
      (hilbertPositiveSegmentMultiple Geo n c)
      (hilbertPositiveSegmentMultiple Geo n d)
      hScaled
case right =>
  exact
    hilbertEudoxus_v14_gt_iff
      Geo
      (hilbertPositiveSegmentMultiple Geo m a)
      (hilbertPositiveSegmentMultiple Geo m b)
      (hilbertPositiveSegmentMultiple Geo n c)
      (hilbertPositiveSegmentMultiple Geo n d)
      hScaled

```

The displayed proof uses only ASCII syntax in the documentation; it is mathematically identical to the production proof.

9 Why V.16 Needs V.14

A tempting approach is to try to prove alternando directly from the original Definition 5 hypothesis

$$a : b = c : d.$$

But Definition 5 compares multiples of a with multiples of b , and multiples of c with multiples of d .

Alternando instead requires comparison of

$$M_m(a) \quad \text{with} \quad M_n(c),$$

and simultaneously

$$M_m(b) \quad \text{with} \quad M_n(d).$$

This is a cross-comparison between the two original ratios.

V.14 is precisely the theorem that converts an equality of ratios into such cross-comparison information.

10 Why V.15 Is the Correct Scaling Step

V.14 can compare corresponding terms only after we have constructed the scaled proportion

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

V.15 supplies the two links

$$a : b = M_m(a) : M_m(b),$$

and

$$c : d = M_n(c) : M_n(d).$$

V.11 then composes these links with the original equality.

Thus the proof is not merely an algebraic rearrangement of symbols. It reconstructs the exact Eudoxean comparison environment needed for arbitrary m, n .

11 Axiomatic Strength

The theorem is stated under

```
[HilbertIncidence Geo]
[HilbertArchimedeanPlane Geo]
```

The Archimedean assumption enters through V.14, whose strict comparison component depends on V.8.

V.11 and V.15 themselves require only the weaker congruence-level magnitude infrastructure.

The assumption profile therefore reflects the deepest dependency in the proposition graph.

12 Relationship with V.11

V.11 appears twice in the proof.

First:

$$M_m(a) : M_m(b) = a : b$$

is composed with

$$a : b = c : d$$

to obtain

$$M_m(a) : M_m(b) = c : d.$$

Second, that result is composed with

$$c : d = M_n(c) : M_n(d)$$

to obtain

$$M_m(a) : M_m(b) = M_n(c) : M_n(d).$$

Thus V.11 is the glue connecting the two V.15 scaling results to the original proportion.

13 Relationship with V.14

Once the fully scaled proportion has been constructed, V.14 finishes the mathematical argument.

No additional order lemma is needed in V.16 itself. The three comparison equivalences returned by V.14 are already exactly the three components of the target Definition 5 statement.

This is an important architectural point: the order theory remains localized inside V.14.

14 Connection with Euclid XI.23

The motivation for formalizing V.16 came from the classical proof of XI.23.

That proof reaches a proportion of the form

$$OL : LM = OP : PQ.$$

V.16 transforms it to

$$OL : OP = LM : PQ.$$

Now a strict comparison such as

$$OP < OL$$

can be transferred to the corresponding comparison

$$PQ < LM.$$

This is the ratio-order step that the earlier proposition-specific XI.23 experiments were reproducing indirectly.

The Book V reconstruction isolates it as reusable mathematics.

15 Two Ratio Languages in the Project

The project already contains another proportion language:

```
HilbertPositiveSegmentProportion
```

based on an angle-coded ratio witness.

V.16 in the present file is instead proved for

```
HilbertEudoxusProportion
```

defined by comparison of all positive multiples.

These two notions are not identified by definition and are not silently interchanged.

A future bridge theorem may establish their equivalence under suitable hypotheses, but V.16 itself belongs entirely to the Eudoxean layer.

16 Hidden Formal Data

Several objects implicit in the classical notation become explicit.

16.1 Arbitrary multiple indices

The proof begins with

```
intro m n
```

because the target proportion is Definition 5.

Thus V.16 is proved pointwise for arbitrary comparison indices.

16.2 Two independent scalings

The first ratio is scaled by m , the second by n . They are not the same multiplier.

This independence is essential: Definition 5 compares arbitrary multiples of the first and third magnitudes.

16.3 Symmetry is explicit

The V.15 statement gives

$$a : b = M_m(a) : M_m(b),$$

whereas the first V.11 composition needs

$$M_m(a) : M_m(b) = a : b.$$

The reversal is therefore performed explicitly by

```
hilbertEudoxusProportion_symm
```

rather than by informal rewriting.

17 Classical DAG versus Formal DAG

The mathematical core is:

$$\begin{array}{c}
a : b = c : d \\
\Downarrow \\
a : b = M_m(a) : M_m(b) \quad c : d = M_n(c) : M_n(d) \\
\Downarrow \\
M_m(a) : M_m(b) = M_n(c) : M_n(d) \\
\Downarrow \\
\text{V.14 comparison transfer} \\
\Downarrow \\
M_m(a) ? M_n(c) \iff M_m(b) ? M_n(d) \\
\Downarrow \\
a : c = b : d.
\end{array}$$

The Lean module graph is:

```

Proposition5_15
Proposition5_11
Proposition5_14
      ↓
Proposition5_16.

```

This is a good example of the project principle that Euclid-numbered propositions should remain visible as separate nodes of the formal dependency graph.

18 Structural Lesson

V.16 vindicates the decision to build a genuine Book V layer rather than to continue adding local ratio helpers inside XI.23.

The theorem is not difficult once the correct infrastructure exists, but its proof draws on several distinct reusable ideas:

- common positive multiples;
- transitivity of ratio equality;

- preservation of comparison across equal ratios;
- Eudoxus' Definition 5.

By separating these into V.15, V.11, V.14, and the shared interface, the formal proof mirrors the mathematical architecture rather than hiding it in one large proposition-specific argument.

19 Conclusion

Proposition V.16 establishes alternando:

$$a : b = c : d \implies a : c = b : d.$$

For arbitrary positive-multiple indices m, n , V.15 scales the two original ratios, V.11 composes the resulting equalities, and V.14 transfers the three possible comparisons between the scaled first terms to the scaled second terms.

Those three equivalences are exactly Definition 5 for the alternated ratio.

This completes the present Book V fragment needed for the classical route back to Euclid XI.23.